VIMAL JYOTHI
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Assignment Cover Page

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PRN VML22CS127	Admission No. 7965
Subject Name :- Data Compression Techniques.	Subject Code:- CST 446
Assignment Title/No : 2	
Name of the faculty: Aiswarya M R	
Assignment Submitted on 20/02/26	

Late submission rules : Max mark will reduced to 50% for 1-5 working day's delay, no mark will be awarded thereafter.

I am hereby confirming that this assignment is my own and I haven't adopted any unfair means in any steps of its preparation to enhance my performance in this assignment.

Date : 20/02/26

Sign with Name
NAMITH.K.P

Assignment subdivision	Maximum Mark	Marks awarded	Remarks
A			
B			
C			

Feed back/suggestions :

Name and sign of the faculty

→ Encoding process of MPEG Layer I

Layer I is the basic MPEG audio compression method.

Step 1: Psychoacoustic Analysis.

* Input Audio is analyzed using psychoacoustic model
* Masking thresholds are calculated.
* Inaudible Components are identified.

Step 2: Subband Division

* Audio signal is divided into frequency bands.
* Based on Critical band principle.

Step 3: Bit Allocation

* More bits assigned to perceptually important bands.
* Fewer bits assigned to masked or less audible bands.

Step 4: Quantization

* Samples are quantized
* Precision reduced where masking allows.

Step 5: Encoding.

* Quantized values Converted into binary format.
* Final compressed bitstream generated.

→ Encoding process of MPEG Layer II

Layer II improved Compression efficiency Compared to Layer I

Step 1: Improved psychoacoustic Analysis.

* more refined masking Calculation
* Better identification of perceptually important signals.

Step 2: Efficient Bit Allocation.

* Bits distributed more efficiently
* Reduced redundancy.

Step 3: Grouped Sample processing.

* Samples grouped together
* Shared parameters reduce overall data size.

Step 4: Quantization

* Quantization applied according to masking threshold.

Step 5: Encoding.

* Quantized samples encoded into digital compressed format.

10. MPEG Audio Compression standards include:

* MPEG Layer I
* MPEG Layer II
* MPEG Layer III (MP3)
* Advanced Audio Coding (AAC)

Unlike traditional signal models, MPEG Audio compression is based on a psychoacoustic model of human hearing, ie, it removes information that cannot be perceived by the human ear.

→ Format of Compressed Audio Data in MPEG

The compressed audio data in mpeg is generated using perceptual coding techniques based on:

* Critical Bands
  - Human ears perceive sounds in frequency ranges called critical bands.
  - These bands are approximately logarithmically spaced.
  - Bandwidth increases with frequency.
  - Used to determine how different frequencies contribute to sound perception.
* Thresholding.
  - Audio components below the thresholds of hearing are removed.
  - Frequencies that are less audible are represented with fewer bits.
  - Helps in reducing data without affecting perceived quality.
* Masking.
  - Masking is a phenomenon where sound makes another sound inaudible.
  * Simultaneous masking
    one sound masks another within the same critical band.
  * Temporal Masking.
    A loud sound mask softer sounds occuring before or after it.

MPEG compression uses masking to discard or reduce masked components.